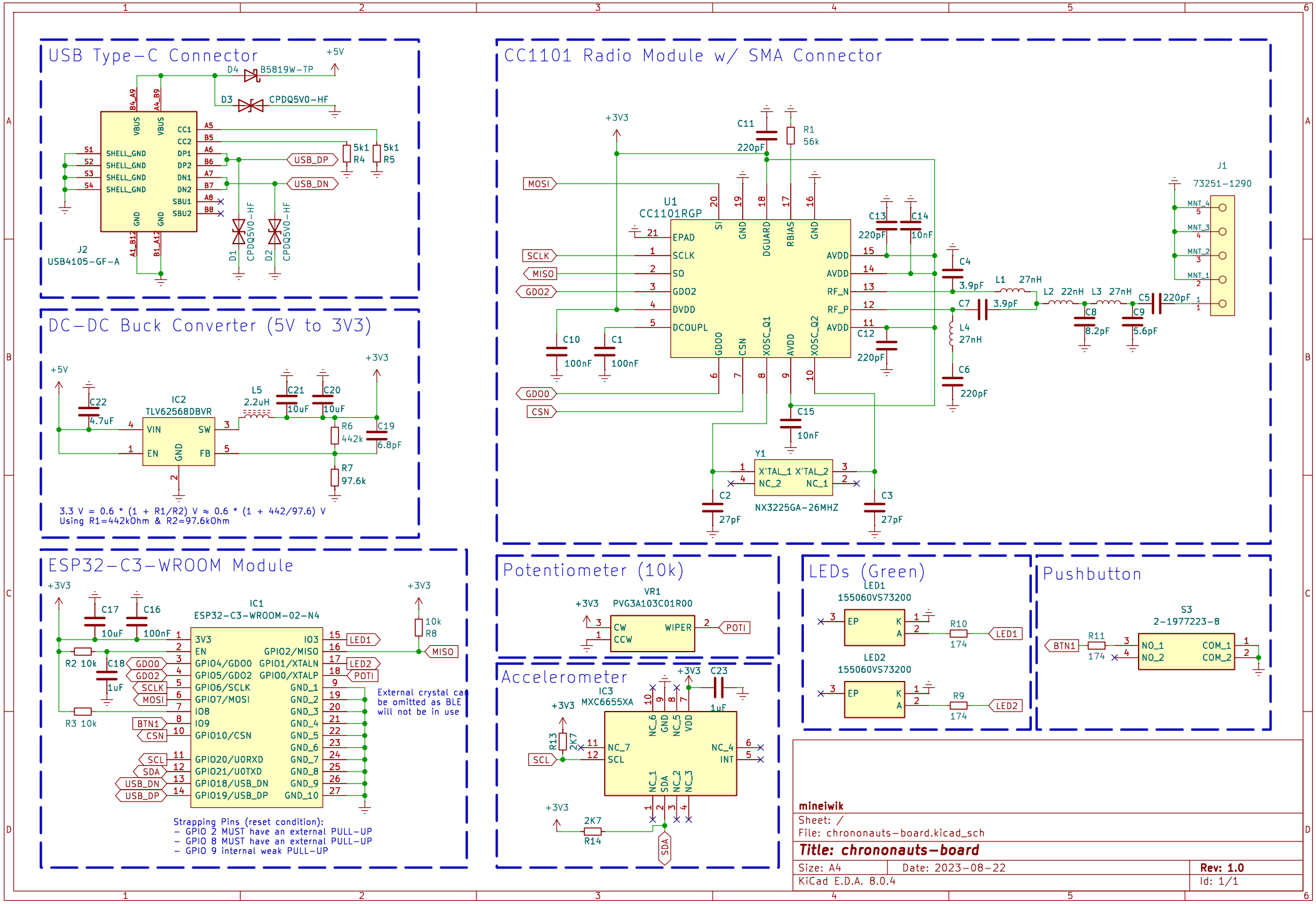
USB Type-C Connector
+5V
D4 B5819W-TP
D3 CPDQ5V0-HF
J2
USB4105-GF-A
VBUS
CC1
CC2
DP1
DP2
DN1
DN2
SBU1
SBU2
SHELL_GND
GND
USB_DP
USB_DN
5k1 R4
5k1 R5
D1 CPDQ5V0-HF
D2 CPDQ5V0-HF
DC-DC Buck Converter (5V to 3V3)
+3V3
IC2
TLV62568DBVR
VIN
SW
EN
GND
FB
C22 4.7uF
L5 2.2uH
C21 10uF
C20 10uF
R6 442k
C19 6.8pF
R7 97.6k
3.3 V = 0.6 * (1 + R1/R2) V ≈ 0.6 * (1 + 442/97.6) V
Using R1=442kOhm & R2=97.6kOhm
CC1101 Radio Module w/ SMA Connector
+3V3
U1
CC1101RGP
EPAD
SCLK
SO
GDO2
DVDD
DCOUPL
GDO0
CSN
SI
GND
DGUARD
RBIAS
AVDD
RF_N
RF_P
XOSC_Q1
XOSC_Q2
MOSI
MISO
C11 220pF
R1 56k
C13 220pF
C14 10nF
C12 220pF
C10 100nF
C1 100nF
C15 10nF
C4 3.9pF
C7 3.9pF
L4 27nH
C6 220pF
L1 27nH
L2 22nH
L3 27nH
C8 8.2pF
C9 5.6pF
C5 220pF
J1
73251-1290
MNT_4
MNT_3
MNT_2
MNT_1
Y1
X'TAL_1
X'TAL_2
NC_2
NC_1
NX3225GA-26MHZ
C2 27pF
C3 27pF
ESP32-C3-WROOM Module
+3V3
IC1
ESP32-C3-WROOM-02-N4
3V3
EN
GPIO4/GDO0
GPIO5/GDO2
GPIO6/SCLK
GPIO7/MOSI
IO8
IO9
GPIO10/CSN
GPIO20/U0RXD
GPIO21/U0TXD
GPIO18/USB_DN
GPIO19/USB_DP
IO3
GPIO2/MISO
GPIO1/XTALN
GPIO0/XTALP
GND_1
GND_2
GND_3
GND_4
GND_5
GND_6
GND_7
GND_8
GND_9
GND_10
C17 10uF
C16 100nF
R2 10k
C18 1uF
R3 10k
BTN1
CSN
SCL
SDA
USB_DN
USB_DP
LED1
LED2
POTI
MISO
10k R8
External crystal can be omitted as BLE will not be in use
Strapping Pins (reset condition):
- GPIO 2 MUST have an external PULL-UP
- GPIO 8 MUST have an external PULL-UP
- GPIO 9 internal weak PULL-UP
Potentiometer (10k)
VR1
PVG3A103C01R00
CW
CCW
WIPER
Accelerometer
IC3
MXC6655XA
NC_7
SCL
NC_4
INT
NC_6
GND
NC_5
VDD
NC_1
SDA
NC_2
NC_3
C23 1uF
R13 2K7
R14 2K7
LEDs (Green)
LED1
155060VS73200
LED2
155060VS73200
EP
K
A
R10 174
R9 174
Pushbutton
S3
2-1977223-8
BTN1
R11 174
NO_1
NO_2
COM_1
COM_2
mineiwik
Sheet: /
File: chrononauts-board.kicad_sch
Title: chrononauts-board
Size: A4
Date: 2023-08-22
Rev: 1.0
KiCad E.D.A. 8.0.4
Id: 1/1